Pilot study on grokking in RL

October 28, 2026

1 Overview

Fernando, David, and I hypothesized that grokking occurs in RL and that it affords better generalization. This is to say that task performance can plateau while continued training leads to more abstract belief representation which then becomes useful when learning a new but related task. Here, we explored this using decoded belief resolution as a proxy for grokking. Whether it is or not was not explored here.

2 Initial Training on The First Task

A single seed is trained, using PPO, to predict the tokens emitted by an HMM. In a second variant, it is trained with PPO and an auxiliary cross entropy next token prediction loss.

The process is the Cyclic Pusher HMM, a three state process with a two-token alphabet, designed by Alec Boyd. Agents have a two-dimensional action space, each corresponding to one of the tokens in the HMM's alphabet. A reward of 1 is given if the emitted action corresponds to the token emitted by the HMM at that time step and 0 otherwise.

The second variant uses an additional head that also has a two-dimensional output on which an NTP cross entropy loss is applied.

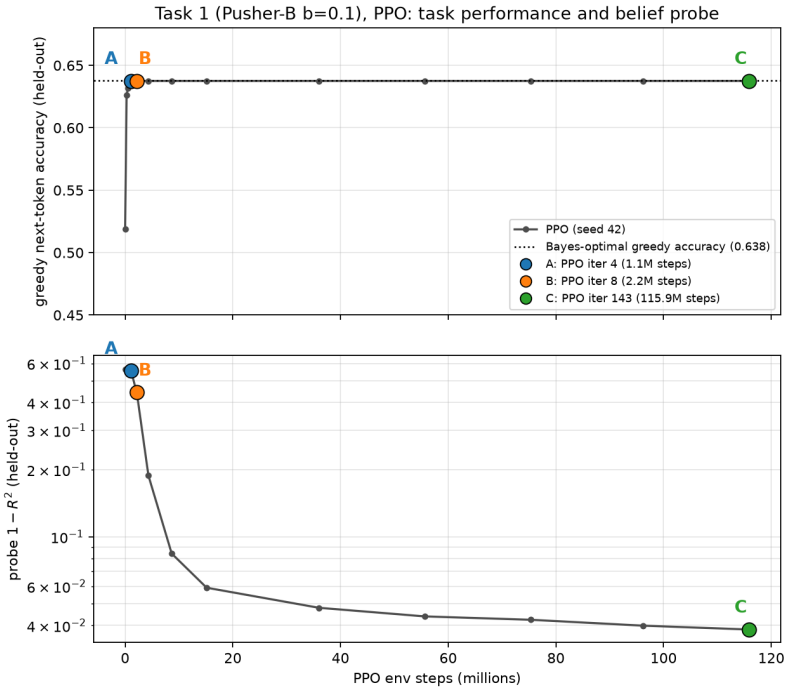
The model's belief representation as measured against the true Bayesian target is measured with a linear probe at each checkpoint.

2.1 Performance and Representation

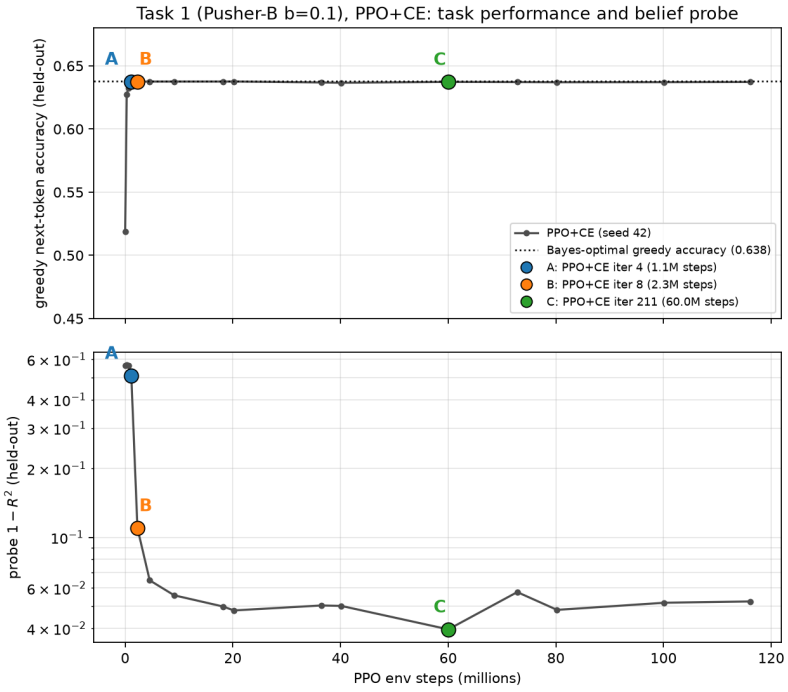
Performance plateaus near optimal while the belief representation improves for a long period after in Figure 1a and 1b. The second variant does not improve convergence time to task competency but does improve representation per training budget.

2.2 Three Important Checkpoints

For each of the two training variants, three checkpoints are highlighted. All three have nearly identical task performance. However, the first checkpoint, A, has much worse belief resolution (as recovered by the linear probe) compared to the third checkpoint, C. The second checkpoint, B, of the PPO-only run also has worse belief resolution but is much improved in the PPO+CE run.



(a) PPO.



(b) PPO+CE.

Figure 1: Held-out greedy accuracy and held-out affine belief-probe $1 - R^2$ on task 1, with selected checkpoints A, B and C marked. Lower $1 - R^2$ means better linear decoding. Each panel shows one training run.

3 The Second Task

Each of the checkpoints are then trained on a new task. The second task is the same as the first but we add a third action, pass, with reward 0, and change the wrong-guess reward to -2; a correct guess still earns $+1$. If the most likely token has conditional probability p, the expected reward of guessing it is $p - 2(1-p) = 3p - 2$. The Bayes rule is therefore to guess when $p > 2/3$, pass when $p < 2/3$, and choose either at equality. Because actions do not change future observations or dynamics, there is no information-gathering advantage to guessing.

3.1 Belief Decodability and Transfer Performance

As predicted, checkpoints with better belief representation in the first task do better on the second task ie C beats A even though all checkpoints have the same performance on the first task ie A, B, and C are equivalent. All checkpoints do better than one trained from scratch (light gray line). Figure 2.

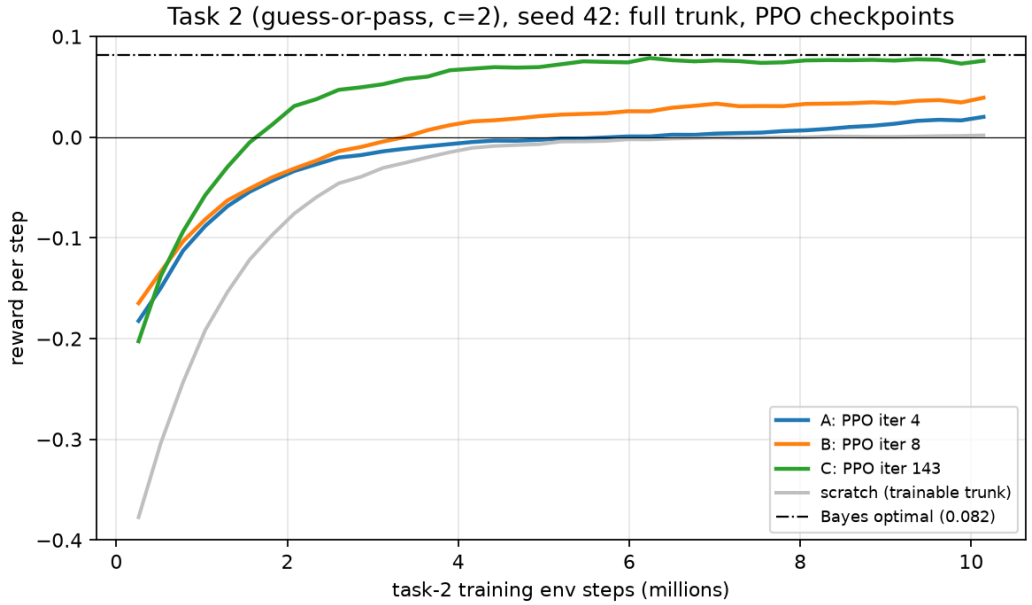


Figure 2: PPO-only full-trunk runs on the second task.

3.2 The Effect of Better Belief Resolution From A Predictive Loss

The clearest observed transfer advantage of PPO+CE over PPO occurs at checkpoint B, where the task-1 belief-probe scores also differ substantially.

Comparing figure 1a to figure 1b shows that belief decoding improves earlier with cross entropy in these runs. At checkpoint B, held-out probe R^2 is 0.556 for PPO and 0.890 for PPO+CE, at similar traning step counts.

Figure 3 compares the performance of PPO-only checkpoints to those from PPO+CE on the second task. Performance on the second task is the same for checkpoints A and C whether or not it was trained with CE. However, checkpoint B shows better second-task performance when it has better belief resolution as conferred by CE.

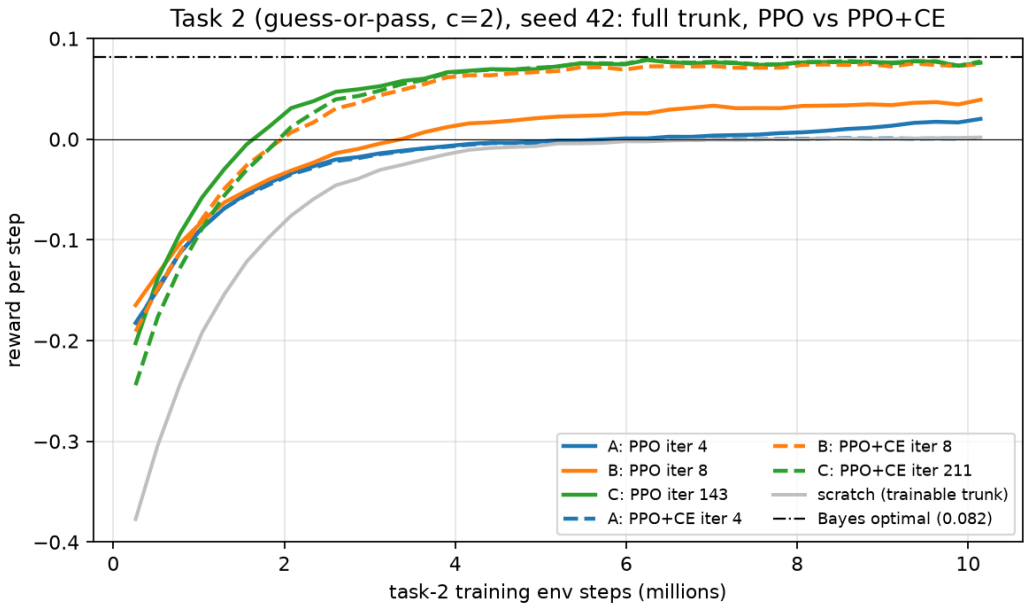


Figure 3: Full-trunk PPO-only runs plus PPO+CE runs on the second task. Checkpoint B is the only one where PPO+CE provides a transfer advantage and is also the only one where the belief representation is better resolved (when comparing checkpoint to checkpoint from PPO-only vs PPO+CE).

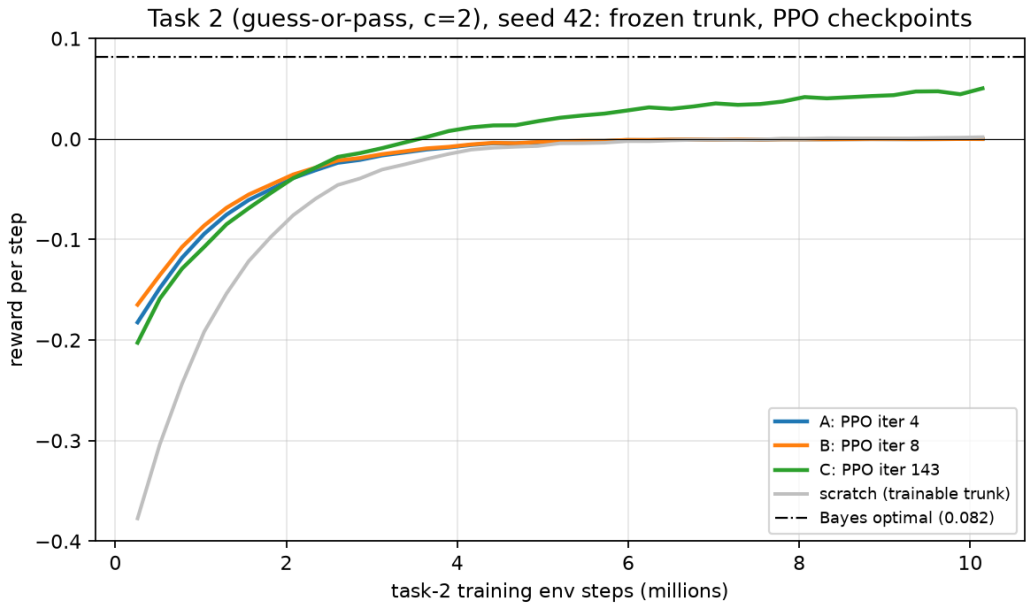
3.3 Freezing The Trunk

Testing on a frozen trunk made learning the second task more difficult but still preserved the ranking of the three checkpoints (Figure 4a and 4b) with the exception of the comparison between PPO-only and PPO+CE on checkpoint B.

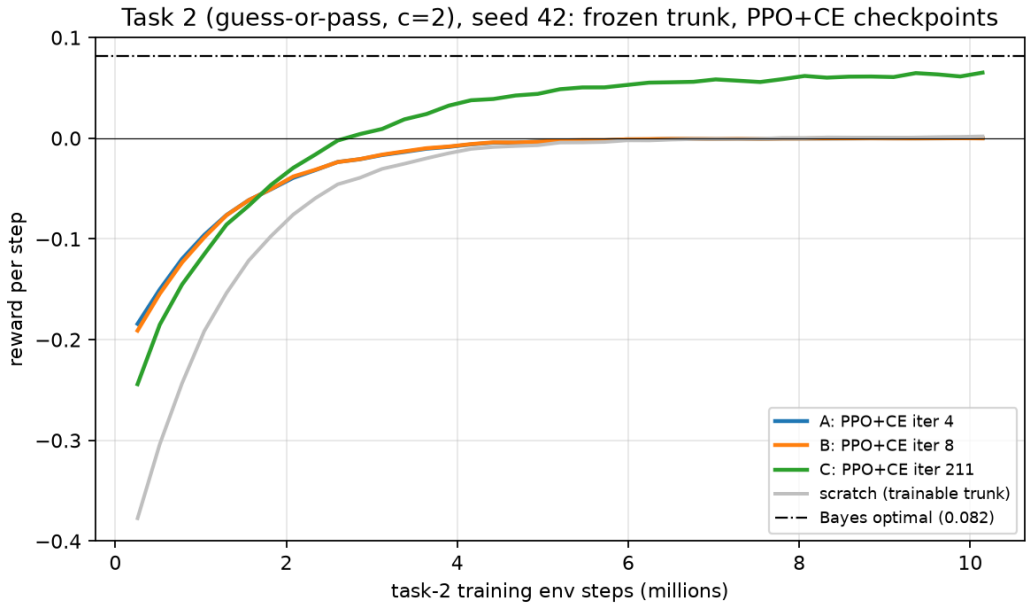
4 Confounds

The later checkpoint performing better on the new task could be the result of factors other than grokking since it has more experience. A better test would be to use off-policy RL on a small-enough set of experience.

Working against the older checkpoints, however, is loss of plasticity typically seen in late-training such as from constant neurons.



(a) PPO-only.



(b) PPO+CE.

Figure 4: Frozen-trunk runs on the second task.